

Neuromechanisms of long-term memory retrieval

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Abstract—Forgetting often occurs due to failures in long-term memory (LTM) retrieval, even when the memory content itself remains intact. Contemporary models of human memory introduces a short-term, or working memory (WM), component that acts as a buffer to LTM. WM performance is known to be influenced by “attention” neuromechanism that selectively filter information necessary for both remembering and later recalling. This raises an intriguing question: is LTM retrieval also modulated by attention, or does it rely on an entirely different neuromechanism? To address this, we devised two behavioral experiments to measure and compare WM and LTM retrieval performance while recalling one or two objects under varying attentional demands. To further identify the neuromechanism behind LTM retrieval, we modeled both WM and LTM with a mixture model to extract two parameters: mnemonic precision (SD) and guessing rate (g), which reflect the utilization of attention and decision-making neuromechanism, respectively. We discovered reversal of trends in mixture model parameters of LTM retrieval data when compared to WM, supporting decision-making as neuromechanism that modulated LTM retrieval.

Index Terms—neuromechanism, long-term memory retrieval, working memory, divided attention, decision-making

I. INTRODUCTION

Human memory consists of multiple interacting systems that enable us to encode, store, and retrieve information. A key distinction exists between working memory (WM), which temporarily maintains information for ongoing tasks, and long-term memory (LTM), which stores information for extended periods. Contemporary models propose that WM acts as a buffer through which information from LTM must pass [6] [17]. While WM is known to be modulated by attention mechanisms that filter and select relevant information [4], the neuromechanisms governing LTM retrieval remain debated.

Both neuroimaging and electrophysiology studies have demonstrated that attention-related neural signals predict WM performance [3] [14]. However, whether similar attentional

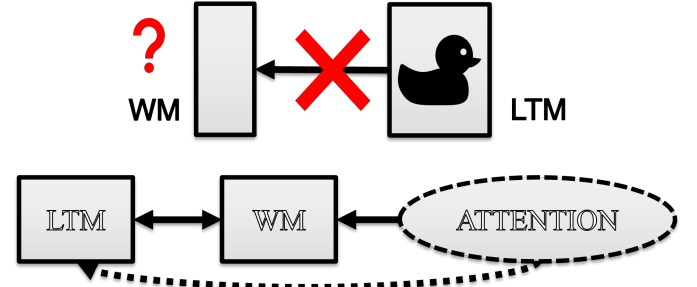


Fig. 1. Conceptual framework showing the relationship between attention, working memory (WM), and long-term memory (LTM). Upper picture depicts a cause of forgetting as retrieval failure from LTM to WM. WM is known to be modulated by attention, but the role of attention in LTM retrieval is unclear.

mechanisms support LTM retrieval, or if LTM relies on distinct neuromechanisms remains unknown. Understanding these mechanisms is crucial for both visual memory research and applications in education and cognitive rehabilitation.

This study aimed to dissociate the roles of attention and decision-making in LTM retrieval through two behavioral experiments comparing WM and LTM performance under varying cognitive loads. We hypothesized that if LTM retrieval shares attentional mechanisms with WM, it should show similar parameter patterns in response to increased cognitive demands.

II. LITERATURE REVIEW

A. Working Memory and Attention

The classical model of working memory proposes a limited capacity of 3-4 independent “slots” for storing visual information [13] [16]. This slot model suggests that each visual item occupies one discrete slot, with no degradation in

precision until the capacity limit is reached. However, recent studies examining the precision of visual working memory have revealed that memory resolution decreases continuously as more items are stored [8] [9].

Recent neuroimaging and electrophysiological studies have shown that alpha-band oscillations (8-12 Hz) track both the maintenance of spatial information in working memory [4] [14] and the retrieval of precise spatial locations from long-term memory [2]. This neural evidence suggests possible shared mechanisms between working memory maintenance and long-term memory retrieval.

B. Mixture and Swap Models

A major advancement in understanding memory mechanisms came through the development of mixture models [8]. These models decompose memory performance into two components:

- A Gaussian distribution centered on the correct target value, reflecting noise in stored representations
- A uniform distribution representing random guessing when items are not maintained

Building on this framework, Bays et al. [9] proposed an enhanced "swap model" that accounts for an additional source of error: incorrectly reporting features of non-target items. This three-component model captures:

- Gaussian variability in stored representations
- Complete storage failures leading to random guessing
- Feature binding errors resulting in swap errors

C. Memory Precision and Resource Allocation

A key debate concerns how memory resources are allocated across multiple items. Some researchers argue for a fixed capacity limit with slots that can be divided among items [13] [16], while others propose a more flexible resource that can be continuously distributed [8] [9]. This has important implications for understanding how attention modulates memory precision and how precision degrades with increasing memory load.

III. METHOD

A. Participants

TABLE I
PARTICIPANT DEMOGRAPHICS

Parameter	Value
Sample Size	17 participants
Male	12 (70.58 %)
Female	5 (29.42 %)
Age Range	18-35 years
Mean Age	20.94 $\pm$ 1.144 years

Seventeen healthy adults from King Mongkut's University of Technology Thonburi participated in the study through convenience sampling from the undergraduate population (see Table I). All participants provided written informed consent

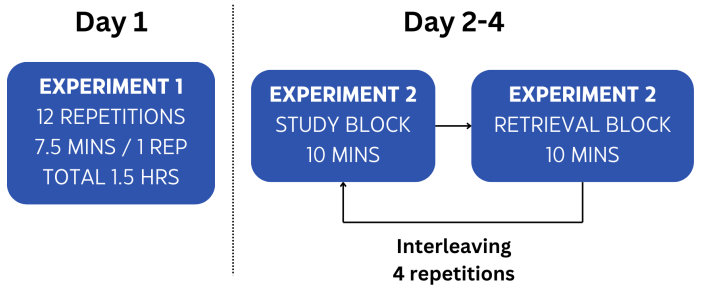


Fig. 2. Experimental procedure across four consecutive days. (A) Day 1: Working memory task. (B) Days 2-4: Long-term memory task which consists of study and retrieval blocks interleaved for 4 repetitions on each session.

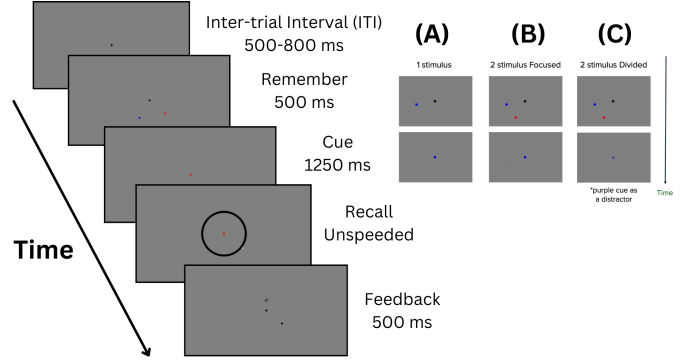


Fig. 3. Working memory task structure. (A) Single stimulus presentation. (B) Two stimuli with focused attention, showing early cue. (C) Two stimuli with divided attention, showing delayed cue.

under institutional review board approval. Three participants were excluded from model-based analysis in long-term memory task due to unreliable fitted model parameters (10 males and 4 females left).

B. Experimental Procedure

The study consisted of four consecutive daily sessions, as shown in Fig. 2. On Day 1, participants completed a 2-hour working memory assessment with breaks provided every 10 minutes. During this session, participants maintained central fixation while responding to stimuli via mouse clicks, receiving trial-by-trial feedback on response accuracy.

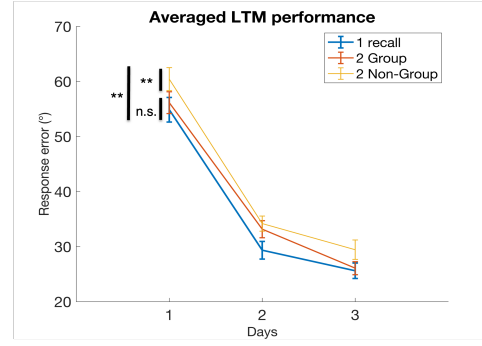
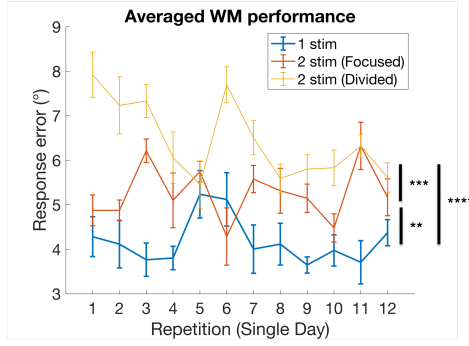
Days 2-4 comprised three consecutive daily sessions of 3 hours each, focused on long-term memory. These sessions included study blocks for learning object-location associations, retrieval blocks for testing memory, and individual retrieval blocks for baseline performance measurement.

We use PsychToolbox [12] for presenting stimuli on 24-inch LCD monitor with refresh rate of 144 Hz.

C. Task Design

1) *Experiment 1: Working Memory Task*: The working memory task employed three conditions to manipulate cognitive load and attentional demands, as illustrated in Fig. 3. In the single stimulus condition, participants remembered one colored dot followed by a brief retention interval and

(A)



(B)

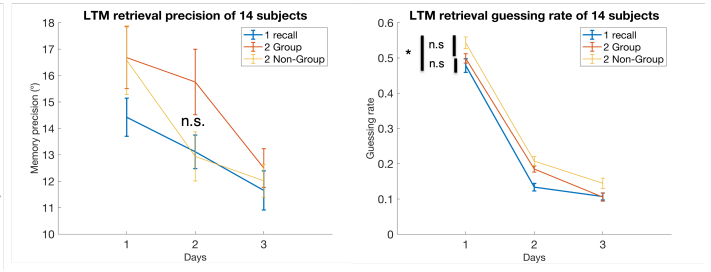
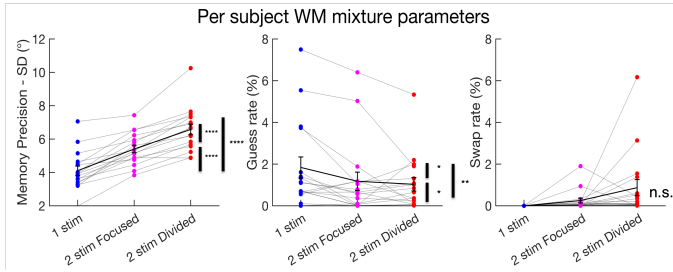


Fig. 4. Performance comparison between working memory and long-term memory tasks. (A) Behavioral accuracy across conditions showing memory performance under conditions that manipulates both cognitive load and attentional demand. (B) Model parameter trends revealing reversal of mnemonic precision and guessing rates trends in LTM when compared to WM. Learning curves demonstrating performance improvements across days. Error bars represent ± 1 SE.

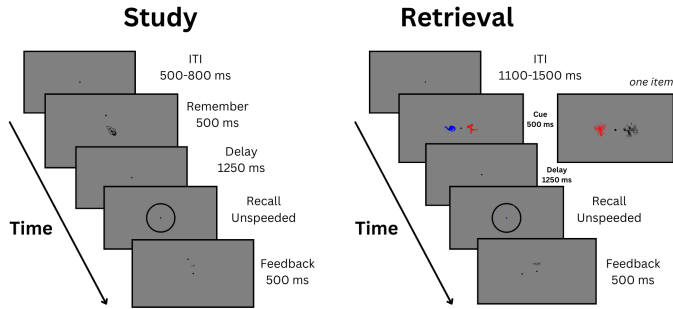


Fig. 5. Long-term memory task structure. Study block showing sequential presentation of object-location pairs. Retrieval block illustrating the two possible cases: one item cue and two items cue that may be grouped or non-grouped.

responded a precise location of presented color cue using a circular response ring. The two stimuli focused attention condition presented two colored dots simultaneously with an early cue indicating the relevant target. The two stimuli divided attention condition similarly presented two dots but used a delayed cue, requiring maintenance of both items in memory. Stimuli geometry and timing follows [5].

2) *Experiment 2: LTM Task*: The long-term memory experiment consisted of three distinct phases, as shown in Fig. 5. During the study phase, participants sequentially viewed 162

object-location pairs with immediate response and feedback for initial encoding. The design eliminated load effects during encoding by presenting items individually. The retrieval phase tested three conditions: single recall with one object-spatial association in which the opposite side will have scrambled placeholder object for balancing visual load, grouped recall in which the object pair will always be presented together on every retrieval block, and non-grouped recall with two objects being randomly paired. The individual retrieval phase provided baseline measurements on all three conditions by separating them into single object presentations to be retrieved sequentially without divided attention requirements to make sure that participants learned all object-location pairs. Stimuli geometry, timing and general experiment procedures follows [2].

D. Data Analysis

1) *Behavioral Measures*: We analyzed two primary outcome measures: response error (deviation from true location, ranging from counterclockwise side -180° to clockwise side $+180^\circ$) and reaction time in seconds. Response errors were calculated as the angular difference between reported and actual locations. We use n-way repeated measures ANOVA and paired t-test for statistical analysis.

2) *Model-based Analysis*: To further identify specific neuromechanisms that underlies LTM retrieval, we use existing

implementation of two computational models to characterize memory performance [11]. The mixture model [8] employed two parameters: guessing rate (g) for random responses in which reflects the use of decision-making mechanisms and mnemonic precision (SD) for target response variability that reflects the use of attention mechanisms. The swap model [9] added a third parameter (B) to account for binding errors in multi-item conditions. Model selection utilized both Corrected Akaike Information Criterion (AICc) and Bayesian Information Criterion (BIC) to assess model parsimony.

IV. RESULTS AND CONCLUSION

A. Results

TABLE II
MODEL COMPARISON METRICS

Metric	Mixture Model	Swap Model
AICc	1020.8	1022.3
BIC	1034.1	1040.1

Two-way ANOVA analysis of working memory performance revealed a significant main effect of condition ($F(2, 16) = [28.9]$, $p < 0.001$), with response errors progressively increasing from single to divided attention conditions. On model selection, one stimulus condition data were fitted with mixture model and two stimuli conditions with swap model to replicate the previous study [3]. The mixture model analysis demonstrated a significant decrease in mnemonic precision (SD) with increased load ($F(2, 16) = [111.09]$, $p < 0.001$) and significant differences in guessing rates across conditions ($F(2, 16) = [4.64]$, $p = 0.017$). Swap rates in two-item conditions were negligible ($F(2, 16) = [2.28]$, $p = 0.15$). Overall, the trends aligned with the recent study in which supporting that WM is being modulated by attention [3], except that there were differences in guessing rates across conditions in which may cause from within-subject perceptual bias due to short cue duration of only 1250 ms [18].

For long-term memory retrieval, a two-way ANOVA revealed significant main effects of both condition ($F(2, 16) = [8.45]$, $p = 0.0011$) and learning across days ($F(2, 16) = [83.31]$, $p < 0.001$). There are significant response error differences that were averaged from all three days span between two items grouped and two items non-grouped conditions [two items grouped: $M = 38.4^\circ$, $SD = 21.9^\circ$; two items non-grouped: $M = 41.3^\circ$, $SD = 24.2^\circ$; $t(16) = -3.87$, $p = 0.0013$, two-tailed] but no differences between one item and two items grouped conditions [one item: $M = 36.6^\circ$, $SD = 23.6^\circ$; two items grouped: $M = 38.4^\circ$, $SD = 21.9^\circ$; $t(16) = -1.3452$, $p = 0.19$, two-tailed], which mean attentional demands have an effect on LTM retrieval performance. Model comparison favored the mixture model (see Table II), with lower AICc (1020.8 vs 1022.3) and BIC (1034.1 vs 1040.1) scores. For mixture model analysis, LTM retrieval showed stable mnemonic precision across conditions ($F(2, 13) = [1.09]$, $p = 0.3527$) but significant changes in guessing rate ($F(2, 13) = [5.64]$, p

$= 0.0092$) which signifies decision-making as an alternative neuromechanism that modulates LTM retrieval under varying attentional demands, also there are clear learning curves on both parameters across days, as shown in Fig. 4, confirming that the trends accurately describing LTM retrieval processes.

B. Conclusion

Our findings provide supporting evidence that long-term memory retrieval operates through decision-making mechanisms that is distinct from the attentional processes governing working memory. The dissociation is most clearly demonstrated by the reversal of parameter trends between tasks: working memory shows classic attention-dependent effects with decreased precision under load and relatively miniscule differences between guessing rates, while long-term memory maintains stable precision but exhibits variable guessing rates with increased cognitive demand.

Limitations of this study is small sample size of only 17 subjects with very specific demographics, which may incur insufficient statistical power for detecting genuine effects.

These results carry theoretical implications for understanding memory organization. Future research direction will be focusing on investigating the neural correlates of these distinct mechanisms and developing targeted interventions for memory enhancement [15]. The differential engagement of attention and decision-making processes suggests that memory training programs might benefit from targeting these mechanisms separately.

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